ECx00x&EG800K&EG810M&EG91xN&EG915K&EG950A Series
FILE Application Note

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
1.0	2020-08-20	Wayne WEI	First official release
1.1	2022-02-25	Wayne WEI	Added the applicable modules EC200A series and EG915N-EU.
1.2	2022-08-05	Chaim QIAN	1. Added the applicable modules EG915N-LA and EG912N-EN. 2. Deleted applicable module EC200T series. 3. Updated the SD card support module to EC200A series (Chapter 1).
1.3	2023-04-21	Amos YAO	1. Added the applicable module EG950A-EL. 2. Added the description of EG950A-EL module supporting SD card (Chapter 1).
1.4	2024-05-22	Amos YAO	1. Updated the applicable modules: ● Added EC200N-LA, EC600M series, EG800K series and EG810M-EU. ● Updated EG950A-EL to EG950A series. 2. Added a note that different models support FILE feature differently due to the flash size of the module (Chapter 1.1).
1.5	2024-12-27	Fei XUE	1. Updated the applicable modules: ● Added EC800K-CN, EC800M-CN and EG915K-EU. ● Deleted EOL products EC200S series and EG912Y-EU. ● Updated EG810M-EU to EG810M series. 2. Updated the declaration of AT command examples (Chapter 2.2). 3. Added the following AT commands: ● AT+QFSIZE (Chapter 2.3.12) ● AT+QFMD5 (Chapter 2.3.13)

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1 Introduction

Quectel ECx00x family, EG800K series, EG810M series, EG91xN family, EG915K-EU and EG950A series modules support AT commands for working with files on different physical storage mediums. This document is a reference guide to these commands.

ECx00x family, EG800K series, EG810M series, EG91xN family, EG915K-EU and EG950A series modules support the following storage mediums:

- **UFS**: User File Storage directory. It is a special directory on the flash file system.
- **SD**: SD card directory. (Only supported by EC200A series and EG950A series).

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00x	EC200A Series
	EC200N-LA
	EC600M Series
	EC800M-CN
	EC800K-CN
-	EG800K Series
-	EG810M Series
EG91xN	EG912N-EN
	EG915N Series
-	EG915K-EU
-	EG950A Series

NOTE

1. The file name will indicate the storage location. When the file name begins with "UFS:", it means that the file is stored in UFS. When the file name begins with “SD:”, it means that the file is stored in SD card. And if there are no prefix characters in the file name, then the file is also stored in UFS.
2. Currently ECx00x family, EG800K series, EG810M series, EG91xN family, EG915K-EU and EG950A series modules do not support the storage medium **RAM**: Random Access Memory.
3. Limited by the flash size of the module, different models of the modules support FILE feature differently. For more details, please contact Quectel Technical Support.

1.2. The Process of Using FILE AT Commands

Apply the following procedure to create, read and write a file in the storage:

1. Upload a file to the storage with **AT+QFUPL**, and output/download it through the serial interface with **AT+QFDWL**.
2. Open the file with **AT+QFOPEN**, and then the file can be written or read at any time and any location until the file is closed with **AT+QFCLOSE**.
   - When opening a file with **AT+QFOPEN**, you can set the file into overwrite mode, read-only mode or other modes with **<mode>** parameter (For more information about **<mode>**, see ***Chapter 2.3.6***). After opening the file, a **<filehandle>** parameter is assigned to it so that various file operations can be carried out.
3. After opening the file, write the data to the file with **AT+QFWRITE** and read the data from the current file position with **AT+QFREAD**.
   - Set the file position with **AT+QFSEEK** or query the current file position with **AT+QFPOSITION**.
4. Close the file with **AT+QFCLOSE**, after which the **<filehandle>** turns invalid.

Commands for managing files in the storage medium:

1. **AT+QFLDS**: Get storage space information.
2. **AT+QFLST**: List the file information in the storage medium.
3. **AT+QFDEL**: Delete the file(s) in the storage medium.

1.3. Description of Data Mode

The COM port of ECx00x family, EG800K series, EG810M series, EG91xN family, EG915K-EU and EG950A series modules has two working modes: AT command mode and data mode. In AT command mode, the inputted data via COM port are treated as AT commands; while in data mode, they are treated as data.

Inputting "**+++**" or pulling up DTR pin (**AT&D1** should be set first) can make the COM port exit data mode. To prevent the "**+++**" from being mistaken for data, the following sequence should be followed:

1) Do not input any character for at least 1 s before inputting "**+++**".
2) Input "**+++**" within 1 s, making sure that no other characters are inputted during that time.
3) Do not input any character within 1 s after inputting "**+++**".

Once **AT+QFUPL**, **AT+QFDWL**, **AT+QFREAD** and **AT+QFWRITE** are executed, the COM port will enter data mode. If you are using "**+++**" or DTR to make the port exit data mode, the executing of these commands will be interrupted before the response is returned. In such a case, the COM port cannot reenter data mode by executing **ATO**.

2 Description of FILE AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Types of AT Commands

Command Type	Syntax	Description
Test Command	**AT+<cmd>=?**	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	**AT+<cmd>?**	Check the current parameter value of the corresponding command.
Write Command	**AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]**	Set user-definable parameter value.
Execution Command	**AT+<cmd>**	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you familiarize with AT commands and learn how to use them. The examples, however, should not be taken as Quectel's recommendation or suggestions about how you should design a program flow or what status you should set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there exists a correlation among these examples and that they should be executed in a given sequence. The URLs, domain names, IP addresses, usernames/accounts, and passwords (if any) in the AT command examples are provided for illustrative and explanatory purposes only, and they should be modified to reflect your actual usage and specific needs.

2.3. AT Command Description

2.3.1. AT+QFLDS Get the Space Information of the Storage Medium

The command gets the space information of the specified storage medium.

AT+QFLDS Get the Space Information of the Storage Medium	
Test Command **AT+QFLDS=?**	Response **OK**
Write Command **AT+QFLDS=<name_pattern>**	Response **+QFLDS: <free_size>,<total_size>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+QFLDS**	Response Return the UFS information **+QFLDS: <UFS_file_size>,<UFS_file_number>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<name_pattern>	String type. Storage medium type. "UFS" UFS "SD" SD card
<free_size>	Integer type. Free space size of **<name_pattern>**.
<total_size>	Integer type. Total space size of **<name_pattern>**.
<UFS_file_size>	Integer type. Size of all files in UFS. Unit: byte.
<UFS_file_number>	Integer type. Number of files in UFS.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

Example

```
AT+QFLDS="UFS"                     //Query the space information of UFS.
+QFLDS: 578847,917503

OK
AT+QFLDS="SD"                      //Query the space information of SD card.
+QFLDS: 251920384,253132800

OK
```

2.3.2. AT+QFLST List the File Information in the Storage Medium

The command lists the information of a single file or all files in the specified storage medium.

AT+QFLST List the File Information in the Storage Medium	
Test Command **AT+QFLST=?**	Response **OK**
Write Command **AT+QFLST=<name_pattern>**	Response **+QFLST: <filename>,<file_size>** **[+QFLST: <filename>,<file_size>** **[…]]** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+QFLST**	Response Return the information of the UFS files: **+QFLST: <filename>,<file_size>** **[+QFLST: <filename>,<file_size>** **[…]]**

	OK If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<name_pattern>	String type. File to be listed.	
	"*"	All the files in UFS
	"**<filename>**"	The specified file **<filename>** in UFS
	"SD:*"	All the files in SD card
	"SD:**<filename>**"	The specified file **<filename>** in SD card
<filename>	String type. File name.	
<file_size>	Integer type. File size. Unit: byte.	
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.	

Example

```
AT+QFLST="*"                          //List all files in UFS.
+QFLST: "UFS:1k.txt",1024
+QFLST: "UFS:2k.txt",2048
+QFLST: "UFS:3k.txt",3072

OK
AT+QFLST="SD:*"                       //List all files in SD card.
+QFLST: "SD:1k.txt",1024
+QFLST: "SD:10K.txt",10240
+QFLST: "SD:100k.txt",102400

OK
```

NOTE

AT+QFLST queries the actual size of the file currently stored in flash. Use **AT+QFWRITE** to write data. If **AT+QFLST** cannot directly query the file size, you need to execute **AT+QFCLOSE** to close the file and then query the file size.

2.3.3. AT+QFDEL Delete the File(s) in the Storage Medium

The command deletes a single file or all the files in the specified storage medium.

AT+QFDEL Delete the File(s) in the Storage Medium	
Test Command **AT+QFDEL=?**	Response **+QFDEL: <filename>** **OK**
Write Command **AT+QFDEL=<filename>**	Response **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filename>	String type. Name of the file to be deleted. Maximum length: 80 bytes.	
	"*"	Delete all files in UFS (do not delete the directory)
	"**<filename>**"	Delete the specified file **<filename>** in UFS
	"SD:*"	Delete all files in SD card (do not delete the directory)
	"SD:**<filename>**"	Delete the specified file **<filename>** in SD card
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.	

Example

```
AT+QFDEL="*"                  //Delete all the files in UFS (do not delete the directory).
OK
AT+QFDEL="UFS:1.txt"          //Delete the 1.txt file in UFS.
OK
AT+QFDEL="SD:*"               //Delete all the files in SD card (do not delete the directory).
OK
```

2.3.4. AT+QFUPL Upload a File to the Storage Medium

The command uploads a file to storage medium. If there is any file in the storage with the same name as the file to be uploaded, an error will be reported.

After executing the Write Command and **CONNECT** returns, the module will switch to data mode. When the uploaded data reach **<file_size>**, or there are no data inputted when **<timeout>** is reached, then it exits data mode automatically. During data transmission, you can use "**+++**" or DTR to make the module exit data mode. For more information, see ***Chapter 1.3***.

AT+QFUPL Upload a File to the Storage Medium	
Test Command **AT+QFUPL=?**	Response **+QFUPL: <filename>[,(1-<freesize>)[,(**range of supported **<timeout>**s**)[,(**list of supported **<ackmode>**s**)]]]** **OK**
Write Command **AT+QFUPL=<filename>[,<file_size>[,<timeout>[,<ackmode>]]]**	Response **CONNECT** TA switches to the data mode (transparent access mode), and the binary data of file can be inputted. When the total size of the inputted data reaches **<file_size>** (unit: byte), TA goes back to command mode and returns the following codes: **+QFUPL: <upload_size>,<checksum>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<freesize>	Integer type. Free space size of **<name_pattern>**. See **AT+QFLDS** for more details of **<name_pattern>**.	
<filename>	String type. Name of the file to be uploaded. Maximum length: 80 bytes.	
	"**<filename>**"	Name of the file to be uploaded to UFS
	"SD:**<filename>**"	Name of the file to be uploaded to SD card
<file_size>	Integer type. File size expected to be uploaded. Default: 10240. Unit: byte.	
<upload_size>	Integer type. Actual size of the uploaded data. Unit: byte.	
<timeout>	Integer type. Time waiting for data to be inputted to USB/UART. Range：1–65535.	

	Default value: 5. Unit: s.
<ackmode>	Integer type. Whether to use ACK mode. 0 Turn off the ACK mode by default 1 Turn on the ACK mode
<checksum>	Integer type. Checksum of the uploaded data.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

NOTE

1. It is strongly recommended to use DOS 8.3 file name format for **<filename>**.
2. **<checksum>** is a 16-bit checksum based on bitwise XOR.
   If the number of the characters is odd, set the last character as the high 8 bit, and the low 8 bit as 0, and then use an XOR operator to calculate the checksum. Inputting **+++** sequence will make the TA to end the data transmission and switch to command mode. However, the data previously uploaded will be preserved in the file.
3. When executing the command, the data must be entered after **CONNECT** is returned.
4. The ACK mode is provided to avoid the loss of data when uploading large files, in case hardware flow control does not work. The ACK mode works as follows:
   1) Run **AT+QFUPL=<filename>,<file_size>,<timeout>,1** to enable the ACK mode.
   2) The module outputs **CONNECT**.
   3) MCU sends 1 KB bytes data, and then module will respond with an **A**.
   4) MCU receives the **A** and then sends the next 1 KB bytes data.
   5) Repeat step 3) and 4) until the transfer is completed.

2.3.5. AT+QFDWL Download a File from the Storage Medium

The command downloads a specified file from the storage medium.

AT+QFDWL Download a File from the Storage Medium	
Test Command **AT+QFDWL=?**	Response **+QFDWL: <filename>** **OK**
Write Command **AT+QFDWL=<filename>**	Response **CONNECT** TA switches to data mode, and the binary data of the file will be outputted. When the content of the file is read, TA goes back to command mode and returns the following codes: **+QFDWL: <download_size>,<checksum>** **OK** If there is an error related to ME functionality:

	+CME ERROR: <err>
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filename>	String type. Name of the file to be downloaded. Maximum length: 80 bytes.	
	"**<filename>**"	Name of the UFS file to be downloaded
	"SD: **<filename>**"	Name of the SD file to be downloaded
<download_size>	Integer type. Size of downloaded data.	
<checksum>	Integer type. Checksum of downloaded data.	
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.	

NOTE

1. Inputting **+++** sequence causes the TA to end the command and switch to command mode.
2. **<checksum>** is a 16-bit checksum based on bitwise XOR.

2.3.6. AT+QFOPEN Open a File

The command opens a file and gets the file handle to be used in commands such as **AT+QFREAD**, **AT+QFWRITE**, **AT+QFSEEK**, **AT+QFPOSITION** and **AT+QFCLOSE**.

AT+QFOPEN Open a File	
Test Command **AT+QFOPEN=?**	Response **+QFOPEN: <filename>[,(**range of supported **<mode>**s**)]** **OK**
Read Command **AT+QFOPEN?**	Response **+QFOPEN: <filename>,<filehandle>,<mode>** **[+QFOPEN: <filename>,<filehandle>,<mode>** **[…]]** **OK**
Write Command **AT+QFOPEN=<filename>[,<mode>]**	Response **+QFOPEN: <filehandle>** **OK** If there is an error related to ME functionality:

	+CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filename>	String type. Name of the file to be opened. Maximum length: 80 bytes. "**<filename>**" Name of the UFS file to be opened "SD: **<filename>**" Name of the SD file to be opened
<filehandle>	Integer type. Handle of the file to be used. The data type is 4 bytes.
<mode>	Integer type. Open mode of the file. <u>0</u> If the file does not exist, it is created. If the file exists, it is opened directly. In both cases, the file can be read and written 1 If the file does not exist, it is created. If the file exists, it is overwritten and cleared. And both of them can be read and written 2 If the file exists, it is opened directly as a read-only file. Otherwise, an error is returned
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

NOTE

If the file is stored in SD card, the obtained **<filehandle>** starts from 20 and increases by 1 by default.

2.3.7. AT+QFREAD Read a File

The command reads the data of a file specified by the file handle. The data start from the current position of the file pointer that belongs to the file handle.

AT+QFREAD Read a File	
Test Command **AT+QFREAD=?**	Response **+QFREAD: <filehandle>[,<length>]** **OK**
Write Command **AT+QFREAD=<filehandle>[,<length>]**	Response **CONNECT <read_length>** TA switches to data mode. When the total size of the data reaches **<length>** (unit: byte), TA goes back to command mode, displays the result and then returns the following codes: **OK**

	If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<length>	Integer type. Length of the file to be read out. Default value: file length. Unit: bytes.
<read_length>	Integer type. Actual read length. Unit: bytes.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

2.3.8. AT+QFWRITE Write a File

The command writes data into a file. The data starts from the current position of the file pointer which belongs to the file handle.

AT+QFWRITE Write a File	
Test Command **AT+QFWRITE=?**	Response **+QFWRITE: <filehandle>[,<length>[,<timeout>]]** **OK**
Write Command **AT+QFWRITE=<filehandle>[,<length>[,<timeout>]]**	Response **CONNECT** TA switches to data mode. When the total size of the written data reaches **<length>** (unit: byte) or the time reaches **<timeout>**, TA goes back to command mode and returns the following codes: **+QFWRITE: <written_length>,<total_length>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<length>	Integer type. Length of the file to be written, and the default length is 10 KB. Maximum value of this parameter is determined by **<freesize>** of **AT+QFUPL**. Unit: bytes.
<timeout>	Integer type. Time waiting for data to be inputted to USB/UART. Default: 5. Unit: s.
<written_length>	Integer type. Actual written length. Unit: bytes.
<total_length>	Integer type. Total file length. Unit: bytes.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

2.3.9. AT+QFSEEK Set a File Pointer to the Specified Position

The command sets a file pointer to the specified position. This will decide the starting position of commands such as **AT+QFREAD**, **AT+QFWRITE** and **AT+QFPOSITION**.

AT+QFSEEK Set a File Pointer to the Specified Position	
Test Command **AT+QFSEEK=?**	Response **+QFSEEK: <filehandle>,<offset>[,<position>]** **OK**
Write Command **AT+QFSEEK=<filehandle>,<offset>[,<position>]**	Response **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<offset>	Integer type. Number of bytes of the file pointer movement.
<position>	Integer type. Pointer movement mode. <u>0</u> The beginning of the file 1 The current position of the pointer 2 The end of the file
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

NOTE

1. If **<position>** is 0 and **<offset>** exceeds file size, the command returns **ERROR**.
2. If **<position>** is 1 and the total size of **<offset>** and the current position of the pointer exceeds the file size, the command returns **ERROR**.
3. If **<position>** is 2, the handle moves forth.

2.3.10. AT+QFPOSITION Get the Offset of a File Pointer

The command gets the offset of a file pointer from the beginning of the file.

AT+QFPOSITION Get the Offset of a File Pointer	
Test Command **AT+QFPOSITION=?**	Response **+QFPOSITION: <filehandle>** **OK**
Write Command **AT+QFPOSITION=<filehandle>**	Response **+QFPOSITION: <offset>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<offset>	Integer type. Offset from the beginning of the file.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

2.3.11. AT+QFCLOSE Close a File

The command closes a file and ends the operation to the file. After that, the file handle is released and should not be used again, unless the file is opened again with **AT+QFOPEN**.

AT+QFCLOSE Close a File	
Test Command **AT+QFCLOSE=?**	Response **+QFCLOSE: <filehandle>** **OK**
Write Command **AT+QFCLOSE=<filehandle>**	Response **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

2.3.12. AT+QFSIZE Query the Size of the Current Open File

This command queries the size of the currently open file.

AT+QFSIZE Query the Size of the Current Open File	
Test Command **AT+QFSIZE=?**	Response **+QFSIZE: <filehandle>** **OK**
Write Command **AT+QFSIZE=<filehandle>**	Response **+QFSIZE: <size>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms

Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filehandle>	Integer type. Handle of the file to be operated.
<size>	Integer type. Size of the current open file.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

2.3.13. AT+QFMD5 Get the MD5 Hash Result of the Current Open File

The command gets the MD5 hash result of the current open file.

AT+QFMD5 Get the MD5 Hash Result of the Current Open File	
Test Command **AT+QFMD5=?**	Response **+QFMD5: <filename>** **OK**
Write Command **AT+QFMD5=<filename>**	Response **+QFMD5: <md5>** **OK** If there is an error related to ME functionality: **+CME ERROR: <err>**
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<filename>	String type. Name of the current open file. Maximum length: 80 bytes. "**<filename>**" Name of the opened UFS file "SD: **<filename>**" Name of the opened SD file
<md5>	Hexadecimal string type. The MD5 hash result of the file.
<err>	Code of an error relating to ME. See ***Chapter 4*** for details.

3 Examples

3.1. Upload and Download a File

3.1.1. Upload a File

3.1.1.1. Non ACK Mode

AT+QFUPL="test1.txt",10	//Upload the text file *test1.txt* to UFS.
CONNECT	
<Input file bin data>	
+QFUPL: 10,3938	
OK	

3.1.1.2. ACK Mode

The ACK mode can make the data transmission more reliable. When transmitting a large file without hardware flow control, the ACK mode is recommended to prevent the data from being lost. For more details about ACK mode, see **AT+QFUPL**.

AT+QFUPL="test.txt",3000,5,1	//Upload the text file *test.txt* to UFS.
CONNECT	
<input file bin data of 1024bytes>	
A	//After receiving 1024 bytes of data, the module returns an **A.** Then the next 1024 bytes of data can be inputted.
<input file bin data of 1024bytes>	
A	
<input the rest file bin data>	
+QFUPL: 3000,B34A	
OK	

3.1.2. Download a File

```
AT+QFDWL="test.txt"                 //Download the text file test.txt from UFS.
CONNECT
<Output Data>
+QFDWL: 10,613e                     //Get the bytes and the checksum value of the uploaded data.

OK
```

3.2. Write and Read a File

3.2.1. Write and Read a UFS File

```
AT+QFOPEN="test.txt",1              //Open the file to get the file handle.
+QFOPEN: 1

OK
AT+QFWRITE=1,10                     //Write 10 bytes to the file.
CONNECT
<Write Data>
+QFWRITE: 10,10                     //The actual bytes written and the size of the file are returned.

OK
AT+QFSEEK=1,0,0                     //Set the file pointer to the beginning of the file.
OK
AT+QFREAD=1,10                      //Read the data.
CONNECT
<Read Data>

OK
AT+QFCLOSE=1                        //Close the file.
OK
```

3.2.2. Write and Read a SD File

```
AT+QFOPEN="SD:1.txt",1              //Open the file to get the file handle.
+QFOPEN: 20

OK
AT+QFWRITE=20,1024                  //Write 10 bytes to the file.
CONNECT
<Write Data>
+QFWRITE: 1024,1024             //The actual bytes written and the size of the file are returned.

OK
AT+QFSEEK=20,0,0                //Set the file pointer to the beginning of the file.
OK
AT+QFREAD=20,1024               //Read the data.
CONNECT
<Read Data>

OK
AT+QFCLOSE=20                   //Close the file.
OK
```

4 Summary of Error Codes

<err> Indicates the error code related to the file operation of the module to which this document applies. See ***Table 3*** for details.

Table 3: Summary of Error Codes

<err>	**Meaning**
400	Invalid input value
401	Larger than the size of the file
402	Read zero byte
403	Drive full
405	File not found
406	Invalid file name
407	File already exists
409	Fail to write the file
410	Fail to open the file
411	Fail to read the file
413	Reach the max number of file allowed to be opened
414	The file read-only
416	Invalid file descriptor
417	Fail to list the file
418	Fail to delete the file
419	Fail to get disk info
420	No space

421	Time out
423	File too large
425	Invalid parameter
426	File already opened

5 Appendix References

Table 4: Terms and Abbreviations

Abbreviation	Description
ACK	Acknowledgement
COM	Communication Port
DOS	Disk Operating System
DTR	Data Terminal Ready
MCU	Microprogrammed Control Unit
ME	Mobile Equipment
RAM	Random Access Memory
SD	Secure Digital
TA	Terminal Adapter
UART	Universal Asynchronous Receiver-Transmitter
UFS	User File Storage
USB	Universal Serial Bus
XOR	Exclusive OR